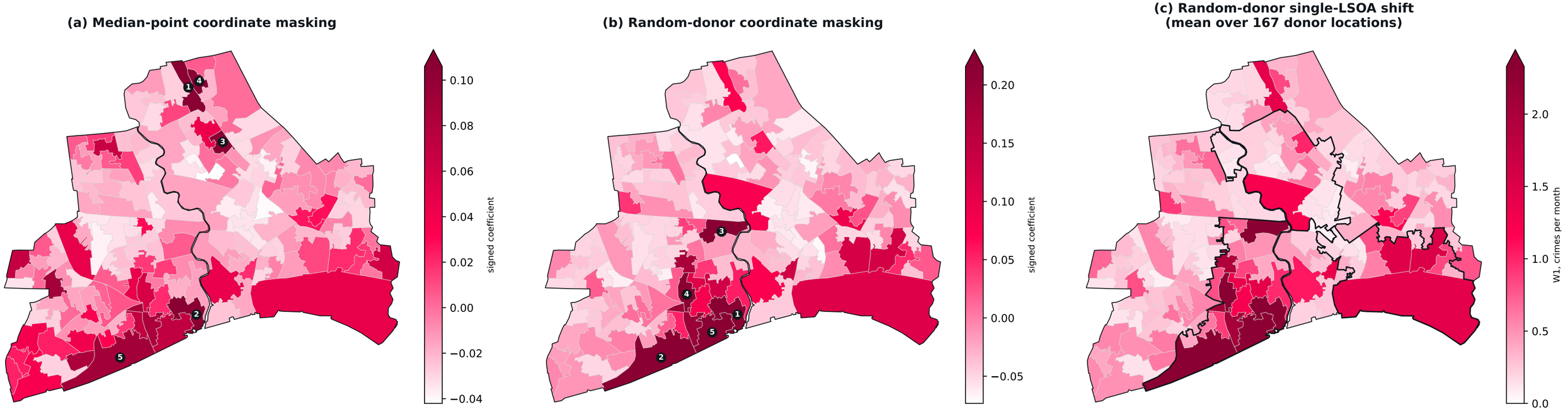


Aligned design (XGBoost on 30 predictors + BNG easting/northing), N=4,000, seed 42. (a) masked coordinates moved to the training-median point; (b, c) moved to random other Hull LSOAs (10 draws per perturbation). Top 5 numbered; (c) black outlines = expert regions. Own sequential scale per panel. Not causal effects.